

stats101a_hw10

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```
library(tidyverse)
```

```
## -- Attaching core tidyverse packages ----- tidyverse 2.0.0 --
## v dplyr      1.2.0      v readr      2.2.0
## v forcats    1.0.1      v stringr   1.6.0
## v ggplot2    4.0.2      v tibble    3.3.1
## v lubridate  1.9.5      v tidyr     1.3.2
## v purrr      1.2.1
## -- Conflicts ----- tidyverse_conflicts() --
## x dplyr::filter() masks stats::filter()
## x dplyr::lag()     masks stats::lag()
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
```

```
df <- data.frame(read.csv('./dietstudy.csv'))
```

```
str(df)
```

```
## 'data.frame':    160 obs. of  45 variables:
## $ SUBJECT : int  1 2 3 4 5 6 7 8 9 10 ...
## $ DIET     : chr  "Atkins" "Atkins" "Atkins" "Atkins" ...
## $ DIETW    : chr  "No" "No" "No" "No" ...
## $ DIETX    : chr  "No" "No" "No" "No" ...
## $ DIETY    : chr  "no" "no" "no" "no" ...
## $ DIETZ    : chr  "yes" "yes" "yes" "yes" ...
## $ AGE      : int  43 23 42 55 66 37 45 53 54 53 ...
## $ SEX      : chr  "Female" "Male" "Male" "Male" ...
## $ WEIGHT_0 : num  92.3 109.5 86.5 118 80.2 ...
## $ BMI_0    : num  36.5 37.9 28.9 31.7 30.9 ...
## $ WAIST_0   : num  123 119 103 110 103 113 109 120 109 111 ...
## $ DROPOUT2 : chr  "no" "no" "no" "no" ...
## $ WEIGHT_2 : num  89.8 104 79.2 115 77.5 ...
## $ BMI_2     : num  35.5 36 26.5 30.9 29.9 ...
## $ WAIST_2   : num  118 112 94.5 108.5 100.5 ...
## $ DROPOUT6 : chr  "no" "no" "no" "no" ...
## $ WEIGHT_6 : num  92 96.2 80.4 117.4 78 ...
## $ BMI_6     : num  36.4 33.3 26.9 31.5 30.1 ...
## $ WAIST_6   : num  121 100.5 95.5 110 96 ...
## $ DROPOU12 : chr  "yes" "no" "no" "no" ...
## $ WEIGH_12 : num  92.3 92.8 79.3 125.5 79.5 ...
## $ BMI_12    : num  36.5 32.1 26.5 33.7 30.7 ...
## $ WAIST_12 : num  123 101.5 94.5 114.5 97.5 ...
```

```
## $ ADHER_1 : int 8 9 8 8 10 9 9 1 10 9 ...
## $ ADHER_2 : int 5 8 7 10 7 9 7 6 10 9 ...
## $ ADHER_3 : int 1 8 6 9 2 7 3 2 10 7 ...
## $ ADHER_4 : int 1 8 7 7 1 4 6 2 9 6 ...
## $ ADHER_5 : int 1 4 4 6 1 3 5 2 9 6 ...
## $ ADHER_6 : int 5 6 2 4 1 1 5 1 7 5 ...
## $ ADHER_7 : int 5 5 3 3 1 3 5 1 7 6 ...
## $ ADHER_8 : int 3 4 2 3 1 1 5 1 7 4 ...
## $ ADHER_9 : int 1 4 5 6 1 2 4 1 5 5 ...
## $ ADHER_10: int 1 2 3 4 1 1 2 1 4 4 ...
## $ ADHER_11: int 1 2 2 4 1 1 3 1 4 3 ...
## $ ADHER_12: int 1 9 8 2 1 1 2 1 7 2 ...
## $ AVEADH_2: num 6.5 8.5 7.5 9 8.5 9 8 3.5 10 9 ...
## $ AVEADH_6: num 3.5 7.17 5.67 7.33 3.67 ...
## $ AVEAD_12: num 2.75 5.75 4.75 5.5 2.33 ...
## $ ADH1212 : num NA 33.06 22.56 30.25 5.44 ...
## $ KCAL_0 : int 1506 2834 3364 1632 1732 1667 1425 1890 2347 1454 ...
## $ KCAL_1 : int 1173 1844 1504 1873 1971 1296 1117 2082 1622 1218 ...
## $ KCAL_2 : int 1506 1411 1640 1802 1935 1981 1027 2082 1585 1372 ...
## $ KCAL_6 : int 1506 1669 2316 2708 1935 1981 1027 1787 1340 1557 ...
## $ AKCAL_6 : int 0 -1165 -1048 1076 203 314 -398 -103 -1007 103 ...
## $ KCAL_12 : int 1506 1669 2316 2370 1613 1981 1189 1787 1866 1019 ...
```

prepare data

```
df <- df |>
  select(DIET, SEX, WEIGHT_0, WEIGH_12, AVEAD_12)
```

```
df <- df |>
  mutate(weight_change = WEIGH_12 - WEIGHT_0)
```

```
df |> slice_sample(n = 2)
```

```
##      DIET      SEX WEIGHT_0 WEIGH_12 AVEAD_12 weight_change
## 1 Ornish Female    97.4      92.6 5.333333         -4.8
## 2 Atkins  Male   111.0     111.0 1.333333          0.0
```

```
df <- df |> select(-WEIGH_12)
```

(Q1)

```
model <- lm(weight_change ~ ., data = df)
summary(model)
```

```
##
## Call:
## lm(formula = weight_change ~ ., data = df)
```

```
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -17.1694  -1.8588  -0.0983   2.8091  13.7366
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)      8.63508    2.68937   3.211  0.00161 **
## DIETOrnish      -0.78623    1.07144  -0.734  0.46419
## DIETWeight Watchers -0.33435    1.07493  -0.311  0.75619
## DIETZone        -0.53511    1.07155  -0.499  0.61823
## SEXMale          0.37143    0.82819   0.448  0.65444
## WEIGHT_0        -0.05624    0.02738  -2.054  0.04169 *
## AVEAD_12        -1.37443    0.16844  -8.160 1.15e-13 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 4.774 on 153 degrees of freedom
## Multiple R-squared:  0.3449, Adjusted R-squared:  0.3192
## F-statistic: 13.42 on 6 and 153 DF,  p-value: 3.378e-12
```

(a)

holding all other variables constant, there is no evidence that diet type does affect weight change. The estimated differences between diets are relatively small relative to standard errors, and the p-values are high (> 0.05), so not statistically significant.

```
reduced <- lm(weight_change ~ SEX + WEIGHT_0 + AVEAD_12, data = df)
anova(reduced, model)
```

```
## Analysis of Variance Table
##
## Model 1: weight_change ~ SEX + WEIGHT_0 + AVEAD_12
## Model 2: weight_change ~ DIET + SEX + WEIGHT_0 + AVEAD_12
##   Res.Df    RSS Df Sum of Sq    F Pr(>F)
## 1     156 3500.8
## 2     153 3487.7  3     13.133 0.192 0.9017
```

Because $p = 0.9017$ which is much larger than 0.05, we fail to reject the null hypothesis H_0 that diet is associated with weight change. There is no evidence that diet type contributes to explaining weight change after accounting for sex, baseline weight, and average adherence

(b)

Yes, the p-value ($1.15e-13$) is far smaller than 0.05, so we reject H_0 and can conclude with very strong evidence that adherence is associated with weight change, controlling for other variables.

The estimated coefficient is -1.374, for each one-unit increase in average adherence, the expected weight change decreases by about 1.374 kg, all other variables held constnat.

(c)

No. Since the p-value for SEXMale is $0.6544 > 0.05$, we fail to reject H_0 – holding all variables constant, there is no statistically significant evidence that males and females differ in average weight change.

(d)

The WEIGHT_0 parameter of -0.056 indicates that for every 1 kg that a person initially started at, they would lose an estimated 0.056 more kg in weight after the 12 months.

(2)

```
model_interaction <- lm(weight_change ~ WEIGHT_0 + SEX * AVEAD_12, data = df)

summary(model_interaction)
```

```
##
## Call:
## lm(formula = weight_change ~ WEIGHT_0 + SEX * AVEAD_12, data = df)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -16.9081  -1.7723  -0.2156   2.6250  13.3387
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)      7.42862    2.67066   2.782  0.00608 **
## WEIGHT_0        -0.05716    0.02695  -2.120  0.03556 *
## SEXMale          1.97418    1.59414   1.238  0.21744
## AVEAD_12        -1.14668    0.25463  -4.503 1.31e-05 ***
## SEXMale:AVEAD_12 -0.39826    0.33269  -1.197  0.23309
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 4.731 on 155 degrees of freedom
## Multiple R-squared:  0.3484, Adjusted R-squared:  0.3316
## F-statistic: 20.72 on 4 and 155 DF, p-value: 1.069e-13
```

Since the p-value of the interaction term is 0.233 (> 0.05), we fail to reject H_0 .

There is no statistically significant evidence that the effect of adherence on weight change differs between males and females.

(3)

```
diets <- df |> mutate(weight0_centered = WEIGHT_0 - mean(WEIGHT_0, na.rm = TRUE)) |> select(-WEIGHT_0)

str(diets)
```

```
## 'data.frame': 160 obs. of 5 variables:
## $ DIET : chr "Atkins" "Atkins" "Atkins" "Atkins" ...
## $ SEX : chr "Female" "Male" "Male" "Male" ...
## $ AVEAD_12 : num 2.75 5.75 4.75 5.5 2.33 ...
## $ weight_change : num 0 -16.7 -7.2 7.5 -0.7 ...
## $ weight0_centered: num -7.32 9.88 -13.12 18.38 -19.42 ...
```

```
model <- lm(weight_change ~ ., data = diets)
```

```
summary(model)
```

```
##
## Call:
## lm(formula = weight_change ~ ., data = diets)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -17.1694  -1.8588  -0.0983   2.8091  13.7366
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)      3.03294     1.06174   2.857  0.00488 **
## DIETOrnish      -0.78623     1.07144  -0.734  0.46419
## DIETWeight Watchers -0.33435     1.07493  -0.311  0.75619
## DIETZone        -0.53511     1.07155  -0.499  0.61823
## SEXMale          0.37143     0.82819   0.448  0.65444
## AVEAD_12        -1.37443     0.16844 -8.160 1.15e-13 ***
## weight0_centered  -0.05624     0.02738  -2.054  0.04169 *
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 4.774 on 153 degrees of freedom
## Multiple R-squared:  0.3449, Adjusted R-squared:  0.3192
## F-statistic: 13.42 on 6 and 153 DF, p-value: 3.378e-12
```

The expected (average) weight change for a female on the Atkins diet whose initial weight is equal to the sample mean initial weight and whose average adherence is 0 would be about 3 kg.